



Paper ID : 261980

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Subject Code:BMBAI205

Roll No:

MBA-INT
(SEM. II) THEORY EXAMINATION 2025-26
BASICS OF STATISTICS

TIME: 3 HRS**M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A**1. Attempt all questions in brief.****02 x 7 = 14**

Q no.	Question	CO	Level
a.	Define Statistics and state its importance in business.	1	K1
b.	Differentiate between primary and secondary data.	1	K2
c.	Calculate Mean of: 10, 20, 30, 40.	2	K3
d.	Define probability and give one example.	3	K1
e.	Explain correlation with suitable example.	4	K2
f.	What is regression analysis?	4	K1
g.	What is index number? State its uses.	5	K2

SECTION B**2. Attempt any three of the following:****07 x 3 = 21**

Q no.	Question	CO	Level
a.	Explain methods of data collection in business statistics.	1	K2
b.	Calculate Arithmetic Mean from grouped data: C.I: 0-10 10-20 20-30 30-40 Freq: 5 8 12 5	2	K3
c.	A card is drawn from a pack of 52 cards. Find probability of getting (i) a red card (ii) a king.	3	K3
d.	Calculate Karl Pearson's coefficient of correlation from given data: X: 10 20 30 40 Y: 15 25 35 45	4	K3
e.	Explain types of index numbers and their uses.	5	K2

SECTION C**3. Attempt any one part of the following:****07 x 1 = 07**

Q no.	Question	CO	Level
a.	Explain the concept of measures of central tendency. Compare Mean, Median and Mode and discuss their suitability in different business situations	2	K4
b.	A manufacturing company records the monthly wages (in ₹ '000) of its workers as follows: C.I: 10-20 20-30 30-40 40-50 50-60 Freq: 5 9 15 11 10 Calculate the Arithmetic Mean using Assumed Mean Method. Interpret the results for wage distribution in the company.	2	K4



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4. Attempt any one part of the following: 07 x 1 = 07

Q no.	Question	CO	Level
a.	Explain the concept of Regression Analysis and its importance in forecasting. Describe the method of constructing regression equations and discuss the differences between correlation and regression with suitable examples.	3	K5
b.	From the following data, find regression equation of Y on X and estimate Y when X = 35: X: 10 20 30 40 Y: 12 22 32 45	4	K4

5. Attempt any one part of the following: 07 x 1 = 07

Q no.	Question	CO	Level
a.	The following data shows advertisement (X) and sales (Y): X: 10 20 30 40 50 Y: 15 25 35 45 60 Calculate Karl Pearson's coefficient of correlation and interpret the result.	4	K4
b.	Explain the concept of Correlation in detail. Discuss different types of correlation and explain Karl Pearson's coefficient of correlation along with its merits and limitations. Also highlight its significance in business decision making.	4	K5

6. Attempt any one part of the following: 07 x 1 = 07

Q no.	Question	CO	Level
a.	Explain the concept of Time Series Analysis. Discuss its components Trend, Seasonal, Cyclical and Irregular variations in detail. Also explain the Additive and Multiplicative models of time series.	5	K5
b.	A company records its annual sales (₹ lakh) as follows: Year: 2018 2019 2020 2021 2022 2023 Sales: 50 55 65 70 75 85 Calculate the Trend values using 3-year Moving Average method.	4	K4

7. Attempt any one part of the following: 07 x 1 = 07

Q no.	Question	CO	Level
a.	Explain Index Numbers and discuss their role in measuring inflation and cost of living.	5	K5
b.	Calculate Laspeyres Price Index from the following data: Commodity: A, B, C P ₀ : 10 20 30 P ₁ : 12 25 35 Q ₀ : 5 4 3 Interpret the result	5	K4